Saarthi Manual

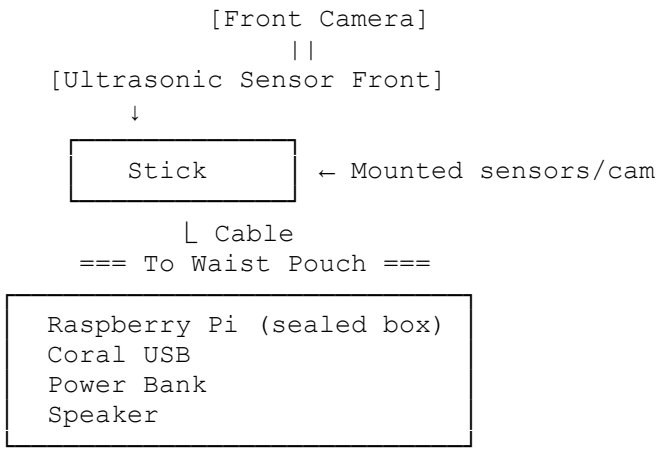
Section 7: Waterproofing, Enclosure Design, and Component Mounting

🎯 7.1 OBJECTIVE

By the end of this section, you will be able to:
✅ Securely mount all components in the stick + pouch setup
✅ Waterproof your electronics without blocking performance
✅ Protect sensors, camera, Pi, Coral USB, and power bank
✅ Ensure comfort, balance, and ease-of-use for the user
✅ Overcome **WeWALK's 2 biggest flaws**:
→ Inability to work in rain
→ No detection of stairs down

🧭 7.2 COMPONENT PLACEMENT OVERVIEW

🎁 Physical Layout (Side View)



🏗️🧰 7.3 MOUNTING EACH COMPONENT

🧭 A. Ultrasonic Sensors (x2)

Sensor	Mounting Position	Angle
Front Sensor	Upper front of stick	90° straight
Downward Sensor	Slightly above ground tip	~45° downward

🔧 Fixing:

- Use a **PVC pipe clamp** or 3D-printed clip

- Add hot glue and waterproof tape around the casing
- Ensure no obstruction in front of sensor (minimum 1.5 cm clearance)

📸 B. Camera (Pi Camera / Night Vision)

| Position: | Middle front of the stick, just above sensors
| Housing: | Use a **transparent acrylic box** or a 3D-printed waterproof frame
| Orientation: | Horizontal, level with user's waist for best object detection range

🔧 Tips:

- Use **a mini acrylic camera enclosure** from Amazon
- Mount with **Velcro + zip ties**, for adjustability
- Route ribbon cable inside a **flexible wire sleeve** toward waist pouch

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🎁 C. Raspberry Pi 5 + Coral USB

| Placement: | Inside waterproof plastic enclosure
| Location: | Inside **waist pouch**, strapped around user's waist
| Heat Management: | Add **air vents + mesh dust filters**, or use **heatsinks**

🔧 Enclosure Box Example:

- SB Components Waterproof ABS Pi Box (₹200–₹300)
- Add silica gel pouch inside to absorb humidity

🔌 D. Power Bank

| Size: | 10,000–20,000 mAh
| Output: | USB-C for Pi (5V/3A), USB-A for Coral
| Placement: | Below or beside the Pi in waist pouch

🔧 Notes:

- Use short **90° angled cables** to reduce tension
- Strap power bank tightly with **Velcro** or foam padding

🔊 E. Speaker / Earphones

| Option 1: Small USB speaker inside pouch
| Option 2: Wired earphones to user's ears
| Option 3 (Recommended): **Bone-conduction open ear headset**

🔧 **Tips:**

- For loud environments, use **USB mini speaker** (~₹250)
- For private use, run audio through **Bluetooth earphones**
- Add **3.5mm extension cable** from Pi to user's shoulder/neck

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☔ 7.4 WATERPROOFING TECHNIQUES

✅ For Raspberry Pi, Coral USB, Power Bank

- Store in **IP65+ rated plastic enclosure**
- Wrap in **two layers**: inside foam, outside pouch
- Add **mini desiccant bags** (silica gel)

✅ For Camera and Sensors

- Use waterproof-specific models (or self-make covers)
- Seal sensor boards with **hot glue on sides + back**
- Use **transparent waterproof tape** over camera (non-reflective)

✅ For Wires

- Wrap exposed ends in **electrical shrink tube**
- Use **nylon braided sleeves** or spiral wraps
- Apply **liquid electrical tape** where necessary

🚿 Rain Test:

✅ Dip just the stick section (not pouch) under flowing water
✅ Check: no false sensor triggers, no camera blur, voice works

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🧵 7.5 HOW TO SECURE CABLES

- Use **spiral cable wrap** or **nylon sleeve** from stick to pouch
- Fix cable at three points along the stick using:
  - **Self-adhesive clips**
  - **Mini zip ties**
  - **Velcro straps**

➡️🧵 Keep cable tension **low**, so movement doesn't affect connections.

🔋 7.6 BALANCE & ERGONOMICS

- Use a **foldable, lightweight cane** (prefer carbon fiber or aluminum)
- Place heavier parts (Pi, power bank) at **waist**, not in stick
- Balance front-mounted sensors **close to grip zone**, not tip
- Add **rubber grip handle** and **foam padding** for comfort

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📏 7.7 PROTOTYPE MEASUREMENTS (Suggested)

Component	Size Estimate (cm)	Weight
Pi + Coral USB	9 × 6 × 3	~150g
Power Bank	12 × 7 × 2	~250g
Camera + Mount	5 × 3 × 2	~50g
Sensors (x2)	4 × 2 × 1 each	~40g
Pouch Weight	—	~500g max load

✅ 7.8 CHECKLIST

✅ Camera clearly sees ahead
✅ Ultrasonic sensors tightly fixed
✅ Pi and power safely sealed
✅ No exposed wires
✅ Comfortable to wear
✅ Easy to remove and debug parts

🔚 SECTION 7 COMPLETE

You've now physically assembled the entire device in a **field-ready**, **weatherproof**, and **modular** form — something even WeWALK doesn't offer.

Saarthi is now:

- Rainproof 🌧️
- Smart 🎯
- Safe and ergonomic 🔋
- Detecting stairs down — a **unique breakthrough**! 🏞️✅

🔜 Next Section:

Section 8 – Testing in Real-World Conditions & Final Adjustments
We'll guide you to:

- Conduct outdoor tests
- Tweak detection sensitivity
- Handle false positives
- Create a testing report + feedback loop